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Promoting social skill use and generalization in children with autism spectrum disorder

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ABSTRACT

Social skills training is a frequently utilized intervention for addressing social deficits of children with autism spectrum disorder (ASD). The current study investigated the effects of the Superheroes Social Skills program, a social skills training program consisting of multiple evidence-based practices, in promoting accurate demonstration of target social skills in both a training and generalized setting. Three children with ASD between the ages of 10 and 14 attended 10 social skills training sessions over five weeks, with social skills lessons targeting participation, conversation, perspective taking, and problem solving skills. A multiple probe design across behaviors and replicated across participants was utilized to evaluate accurate demonstration of target social skills. Results suggest improvement in skill accuracy in both the training and generalization settings. Indirect secondary measures of participant social functioning and parental stress were also collected and suggest improvements associated with social skills training.

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Autism spectrum disorder (ASD) is characterized by significant impairments in social communication and social interaction across contexts (American Psychological Association, 2013). The social impairments experienced by children with ASD are significant and have been described as one of the most salient features of ASD (Carter, Davis, Kiln, & Volkmar, 2005). In order to address common social deficits experienced by children with ASD, social skills training programs are the most commonly implemented intervention (Goin-Kochel, Myers, & Mackintosh, 2007). Social skills training for children with ASD typically includes teaching discrete social skills through observation, modeling, rehearsal, and feedback (Elliott, Roach, & Beddow, 2008).

A number of social skills training strategies have been found to be useful in promoting discrete social skill acquisition and use in training settings by children with ASD. For example, Plavnick, Sam, Hume, and Odom (2013) found a video modeling social skills intervention to produce substantial improvements in complex initiation, awareness, and social reciprocity behaviors of four adolescents with ASD. Utilization of a strength-based social skills intervention in which individual interests

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of children with ASD were incorporated into a Power Card (Gagnon, 2001), was found to result in increases in direction following of children with ASD (Campbell & Tincani, 2011). Doggett, Krasno, Koegel, and Koegel (2013) found a self-management intervention to be effective in increasing questions used in context of two children with ASD. Implementation of a recess-based peer-mediated intervention resulted in increases in communicative acts of children with ASD (Miller et al., 2014). Research has also found peer-mediated interventions incorporating visual scripts to produce improvements in discrete communicative behaviors, such as asking questions and requesting assistance (Ganz et al., 2012).

Although children with ASD are often found to benefit from social skills interventions in training environments, and parent and participant satisfaction with interventions is often high (Cotugno, 2009), research suggests that skills are not often generalized to novel settings or persons (e,g, Bellini, Peters, Brenner, & Hopf, 2007; Ganz et al., 2012; Rao, Beidel, & Murray, 2008; Yakubova & Taber-Doughty, 2013). As such, generalization has been identified as one of the most challenging aspects of social skills training for children with ASD (Barry, Klinger, Lee, Palardy, Gilmore, & Bodin, 2003). The failure of social skills training to produce generalized effects suggests that many social skills interventions do not result in increased social competence, defined by the ability to utilize basic and specific social skills in the appropriate context (Gresham & Elliott, 1987). In order to promote lasting improvements in social competence, it is essential that explicit methodological programming for generalization occur within social skills training.

Stokes and Osnes (1989) defined three generalization promoting categories, which when applied to social skills training for children with ASD, may be effective in overcoming barriers to generalization: exploiting current functional contingencies of reinforcement; training diversely; and engaging functional mediators of behavior. Exploitation of current functional contingencies describes the process of allowing natural consequences of behaviors to maintain behaviors after intervention procedures have been withdrawn (i.e., providing natural consequences for social skill use). Training diversely through incorporation of multiple stimulus and response exemplars ensures that the social skills participant has been exposed to sufficient behavioral examples to be able to respond appropriately to a range of social situations. Training diversely may also be accomplished through varying presentation of training situations. Incorporation of functional mediators of behavior describes the use of strategies that enable self-control of target social skills. Self-monitoring and evaluation of social skill use, training with peers who will later cue performance of a skill, or incorporation of other common salient social stimuli engage functional mediators of social skills use, enhancing the probability that skills are generalized.

The Superheroes Social Skills program (Jenson et al., 2011) was developed to overcome poor generalization effects of social skills training (e.g., Bellini et al., 2007) through incorporation of multiple strategies that address Stokes and Osnes (1989) categories of generalization technologies. Exploitation of current functional contingencies is addressed through utilization of peers in the training process, increasing the likelihood that participants with ASD will receive natural reinforcement from peers for social skill use (e.g., access to tangibles or activities, praise from peers). Superheroes Social Skills trains diversely through exposing participants to multiple stimulus and response exemplars presented through video models of similarly aged peers, role-play scenarios, and social narrative comic books. Functional mediators of behavior are incorporated into the program through encouraging participants to self-monitor target social behaviors utilized within and outside of the training environment, and through incorporation of peers who may serve as cues for social skill use.

The Superheroes Social Skills program has been found to be effective in increasing social engagement duration of children with ASD in training and generalized environments. Radley, Jenson, Clark, & O'Neill (2014) utilized a multiple baseline across participants design to train five parents of children with ASD to facilitate lessons from the Superheroes Social Skills curriculum in a clinic setting. Following parent-facilitated training, children with ASD were introduced to a free-play setting in which peers were available for interaction. Results indicated improvements in duration of social engagements of participants with peers following the introduction of social skills training, with calculation of effect size indicating a moderate mean effect (ES = 0.64). Radley et al. also found parent-facilitated social skills to result in decreases in parental reports of stress. Although improvements in engagement behaviors were observed, no direct measure of skill use or generalization was utilized. In another study, Radley, Ford, Battaglia, and McHugh (in press) utilized a multiple baseline across participants design to evaluate the effects of school-based implementation of the Superheroes Social Skills program on social engagement behaviors of four participants with ASD. Training was found to result in substantial improvements in duration of social engagement of participants with ASD with peers during recess periods (NAP = 0.97). Additionally, analysis of pre- and post-intervention checklists found significant improvements in parent-reports of social skill use. However, as in Radley et al. (2014), no direct measure of target skill accuracy in training or generalized settings was utilized. Given limitations of previous research, it is presently unknown whether the use of the Superheroes Social Skills program would result in mastery of targeted social skills in training or generalized contexts.

The purpose of the present study was to expand on the previous literature through testing the efficacy of the Superheroes Social Skills program in promoting accurate demonstration of target social skills in both training and generalized settings. Although previous studies have examined the utility of the Superheroes Social Skills program in increasing the duration of social engagement of participants with ASD, the present study is the first to evaluate the effect of participation on accurate demonstration of targeted social skills. It was hypothesized that participation in social skills training would result in improvements in accurate demonstration of target social skills. It was also hypothesized that participation in the intervention would result in improved social functioning and reductions in parental stress, as reported by parents.

1. Method

1.1. Participants and settings

Participants included three children with ASD. The participants, each having a prior diagnosis of ASD, were referred to a university-based clinic for social skills training. Prior to inclusion in the study, parents and guardians were given a detailed review of the purpose and procedures of the social skills group, and parental consent for participation was obtained. Cameron, a 10-year, 7-month old Caucasian male, was diagnosed with Asperger's Syndrome by a licensed psychologist. Cameron did not receive supplemental special education services at his school but was placed in his school's gifted program. Sloane, a 14-year, 7-month old Caucasian female, was diagnosed with autism, anxiety disorder, and obsessive-compulsive disorder by a licensed psychologist. At the time of the study, Sloane was prescribed Straterra, Zoloft, and Klonopin. Sloane did not receive supplemental special education services at her school, but had previously received school-based speech-language services and clinic-based services for anxiety-related symptoms. Ferris, an 11-year, 8-month old African-American male, had been previously diagnosed with autism and mild intellectual disability by a licensed psychologist. Ferris received extensive special education services, and was placed in a self-contained classroom at the time of the study. Ferris had previously received clinic-based services for encopresis and aggressive behaviors in home and school settings.

Social skills training and probes of skill accuracy in the training setting took place in a clinic-based treatment room, approximately 3 m by 6 m in size. A table and chairs were placed at the center of the room, parallel to a television that was used to display instructional videos. Assessment of skill generalization across persons, settings, and stimuli was conducted in a nearby room, approximately 3 m by 3 m, in which a table, chairs, and couch were present.

1.2. Materials

Materials utilized in the current study include the Superheroes Social Skills social narrative comics, self-monitoring cards, and videos of the four skills trained in this study (i.e., Participation, Conversation, Perspective Taking, and Problem Solving). The videos were played on a laptop computer, which the researchers displayed on a television monitor for more convenient viewing. Other materials utilized in the current study included games and toys (i.e., Connect Four, Jenga, Transformers action figures, Hot Wheels cars and racetrack, SpongeBob Ants in the Pants, and a large bouncy-ball), used to probe skill accuracy.

1.3. Measures

1.3.1. Skill probes in training setting

Demonstration of target skills in the training setting was a primary dependent variable of the current study. To assess accurate skill use, two graduate student researchers engaged participants with games, toys, and dialog, providing cues for the use of target social skills (e.g. "Come play Jenga with me," "Tell me what you did this weekend," "Show me what you would do if I fell and was holding my knee," "Show me what you would do if someone stepped on your new shoes"). Using task analyses adapted from the Superheroes Social Skills manual (Table 1), the graduate student researchers recorded the skill steps correctly demonstrated by the participant and calculated the percentage of steps correctly demonstrated. For each skill assessed in the training setting, three probes were delivered per session.

1.3.2. Generalized skill probes

Generalization of target skills across persons, settings, and stimuli was also a primary dependent variable. Probes of generalization took place in a nearby room and were administered by three novel graduate student researchers. The novel

Table 1
Task analyses of target social skills.

Participation	Conversation	Perspective Taking	Problem Solving
1. Demonstrate close physical proximity to interaction partner (within 1 meter)	1. Within 3-s of another person beginning a conversation, orient whole body toward speaker	1. Initiate and sustain eye contact (3-s) with person	1. Demonstrate relaxed posture (e.g., relaxed hands and arms) and even paced tone of voice
2. Initiate and sustain eye contact with activity partner (3-s)	2. Initiate and sustain eye contact with conversational partner (3-s)	2. Ask how the person feels	2. Verbally summarize the presenting problem
3. Wait for turn without skipping others or interrupting turn taking progression	3. Demonstrate listening by sustaining eye contact while other person is speaking (3-s)	3. Demonstrate listening by sustaining eye contact (3-s) and head nodding	3. Verbally identify at least two possible solutions to the problem
4. Join in without disrupting the progression of activity using an activity-appropriate response	4. Wait for turn without talking over or interrupting the speaker	4. Ask conversational partner how you can help	4. Identify the better of the available strategies
	5. Make a comment or ask a question related to conversation		5. Utilize the chosen strategy

graduate student researchers delivered three probes per skill using cues different from those used in the training setting (e.g., "Come read a book with us," "Tell me about a movie you like," "Show me what you would do if I were crying," "Show me what you would do if someone bumped you in the hall"). Using the task analyses presented in Table 1, the researchers determined whether each step of the target social skill was completed correctly following presentation of the cues and calculated the percentage of correctly demonstrated steps.

1.3.3. Social functioning

Participant's parents and guardians completed the Autism Social Skills Profile (ASSP, Bellini & Hopf, 2007) to identify social skills to be targeted during intervention. The ASSP was readministered upon conclusion of intervention to evaluate changes in parent report of social functioning. Completion of the ASSP results in a Total Social Functioning score and three subscores: Social Reciprocity; Detrimental Social Behaviors; and Participation/Avoidance. Samples of items on the ASSP include "initiates greetings with others," "joins in activities with peers," "and "makes inappropriate comments." Items on the ASSP are endorsed on a 4-point Likert scale, with rankings ranging from *never* (1) to *very often* (4). High scores on ASSP scales suggest that the child engages in more socially appropriate behavior. Technical evaluations of the ASSP have found high internal consistency ($\alpha = .94$) and test–retest reliability was found to be .90 (Bellini & Hopf, 2007).

1.3.4. Parental stress

Parents and guardians of participants completed the Parenting Stress Index: Short Form (PSI/SF; Abidin, 1995) prior to and immediately following conclusion of social skills training to evaluate changes in parent stress associated with introduction of social skills training. Completion of the PSI/SF results in a Total Stress score and three subscale scores: Parental Distress (PD); Parent–Child Dysfunctional Interaction (P-CDI); and Difficult Child (DC). Evaluations of the PSI/SF have found high internal consistency ($\alpha = .91$), and test-retest reliability was found to be .84 (Abidin).

1.3.5. Social validity

The Behavior Intervention Rating Scale (BIRS; Elliott & Treuting, 1991) was utilized to assess parent perceptions of acceptability and utility of the social skills intervention. The BIRS consists of 24 items rated on a 6-point Likert scale, with rankings from *strongly disagree* (1) to *strongly agree* (6). High scores on the BIRS indicate high levels of participant satisfaction with the intervention. Technical evaluations of the BIRS have found a high alpha coefficient ($\alpha = .97$) for parent completed BIRS (Elliott & Treuting, 1991).

1.4. Experimental conditions

1.4.1. Design

A multiple probe across behaviors with concurrent replication across participants design (Gast, 2010) was utilized to assess the efficacy of the Superheroes Social Skills program in promoting accurate demonstration of target social skills. The current study consisted of three experimental phases: baseline, intervention, and maintenance. Phase changes (i.e., introduction of direct instruction of new target skills) were made when all participants demonstrated a minimum of two consecutive probes of 100% for the skill in the training setting. This criterion was selected as performance less than 100% would not suggest clinically meaningful use of the target skill.

1.4.2. Baseline

Prior to collection of baseline data, parents and guardians of participants were provided with the ASSP and the PSI. Both checklists were completed independently at home and returned for scoring. Results of the ASSP were utilized to identify social skills endorsed as deficits across all participants, with parents and guardians endorsing participation skills, conversation skills, taking others' perspective, and problem solving as salient social deficits. Following identification of target social skills, baseline skill probes were conducted in both the training and generalization settings. During baseline skill probes, praise was provided for attending to tasks. However, no performance feedback was provided to participants. Phase change decisions were made dependent upon trend and stability of data.

1.4.3. Intervention

Social skills training took place during 10 1.5–2 h sessions, conducted over five weeks. Two graduate student researchers facilitated the social skills lessons. Four skills, selected for inclusion due to correspondence with parent-endorsed social skill deficits on the ASSP, were taught: (1), Participation; (2), Conversation; (3), Perspective Taking; and (4), Problem Solving. Social skills were presented in a group format, with all participants receiving instruction in the same target skill.

The procedures utilized during the intervention phase followed the Superheroes Social Skills manual (Jenson et al., 2011). At the beginning of each intervention session, the facilitators probed accurate skill demonstration of target skills in baseline and/or maintenance phases. The skill in intervention was then presented via the Superheroes Social Skills DVD: animated superheroes presented the skill and provided rationale for skill use; the superheroes presented four to five discrete steps for

target skill demonstration; and video models of unknown peers successfully demonstrating the skill were viewed. After viewing the DVD, facilitators reviewed the skill steps with participants and modeled incorrect and correct skill usage. During modeling, participants were encouraged to identify whether the skill steps had been correctly demonstrated and error correction was provided. Participants were then provided with self-monitoring cards and directed to engage in three to four role-plays with the facilitators and other participants. During role-plays, facilitators provided performance feedback should a participant fail to correctly demonstrate all steps to the target skill, and encouraged participants to record successful demonstrations of the target skill on a self-monitoring card. In addition, participants received behavior-specific praise for successful skill demonstration during role-plays. Role-plays were filmed and sent to parents and guardians via email to be reviewed at home.

Following role-plays, skill probes in the training setting for the recently trained skill were administered using procedures previously described. Behavior-specific praise was provided for accurate skill demonstration. Upon completion of probes in the training setting, participants were given 15 min of free play. During this time, participants had access to a variety of games and toys. Probes of skill generalization across persons, settings, and stimuli were administered by novel graduate student researchers in a nearby room upon the conclusion of the 15 min free play period. Behavior-specific praise was provided for accurate skill demonstration in the generalization setting, and sequential modification (Stokes & Baer, 1977), in which performance feedback is provided when generalization is not observed, was utilized following a probe in which a participant demonstrated inaccurate skill use. After generalization probes were completed, participants were returned to the training setting. An animated social narrative comic book in which the superheroes reviewed use of the target skill was then viewed via DVD. Participants were also provided with a paper copy of the social narrative comic book. At the end of each social skills session, the participants were provided with a reward (i.e., candy or crackers) for participation throughout the session. Finally, the facilitators encouraged participants to review videos recordings of role-plays with their parent, practice the target skill at home, and read a copy of the social narrative comic book before the next session. After all participants demonstrated a minimum of two consecutive skill probes of 100% in the training setting, a new target skill was introduced.

1.4.4. Maintenance

Beginning approximately five days after reaching mastery criterion, previously taught skills were probed in both the training and generalization settings to assess for skill maintenance. Maintenance probes were collected at the beginning of social skills sessions. Maintenance probes were collected using procedures identical to those described for baseline probes, except that accurate skill demonstration was followed by behavior-specific praise. Similar to probes collected during the baseline phase, three probes were presented for each skill assessed during each session. Researchers recorded the skill steps correctly demonstrated during each probe, and the percentage of correctly demonstrated steps was calculated.

1.5. Interobserver agreement

To calculate interobserver agreement (IOA), a graduate student researcher who did not deliver skill probes in either setting coded skill acquisition independently and simultaneously with the primary observer. IOA was obtained during 41% of baseline and 36% of intervention and maintenance skill probes in both training and generalization settings. To calculate IOA, the number of agreements in steps completed was divided by the total number of agreements and disagreements and multiplied by 100. Mean overall IOA was calculated to be 96%. Individually, mean IOA for Cameron was calculated at 96% (range, 83–100%), Sloane at 98% (range, 83–100%), and Ferris at 96% (range, 83–100%). Additionally, Kappa was calculated using the formula provided by Uebersax (1982) and was found to be 0.92 (95% CI = 0.91–0.94).

1.6. Treatment integrity

Treatment integrity was assessed using an intervention manual-derived implementation checklist (Appendix A). During the intervention phase, checklists were completed by the two group facilitators during social skills training. Treatment integrity checklists were completed during 100% of intervention sessions. Evaluation yielded 100% integrity for all sessions, with IOA for integrity found to also be 100%.

1.7. Data analysis

Effects of the Superheroes Social Skills program on accurate demonstration of target social skills were evaluated through visual analysis of level, trend, and variability of the data. Additionally, nonoverlap of all pairs (NAP) was calculated to determine magnitude of intervention effect. NAP is a nonparametric method for determining data overlap between each baseline data point and each intervention data point. NAP was calculated using procedures described by Parker and Vannest (2009), but it is also obtained as the Area Under the Curve (AUC) from a Receiver Operator Characteristics (ROC) test module. NAP is strongly correlated with the effect size R^2, making it particularly useful in evaluation of single-case research (Parker & Vannest, 2009). According to Parker and Vannest, NAP scores 0.00–0.65 should be considered weak effects, scores from 0.66 to 0.91 moderate effects, and 0.92–1.00 should be considered

strong effects. Finally, paired samples *t*-tests were used to evaluate the pre- to post-intervention changes from the ASSP and PSI.

2. Results

2.1. Probes of target skills

The primary dependent variable of the current study was the accurate demonstration of target skills in training and generalized settings. Cameron demonstrated mean skill accuracy in baseline that ranged from 10% to 68% in the training setting and from 7% to 60% in the generalization setting (Fig. 1). Following introduction of direct training, immediate increases in level and trend and reductions in variability were observed for target skills in both training and generalization probes. Mean levels of accuracy during intervention ranged from 96% to 100% for training probes and from 53% to 97% for generalization probes. Probes conducted in the training setting revealed maintenance of the Participation, Conversation, and Perspective Taking skills at levels greater than baseline. Maintenance at or above intervention levels was observed for the Participation and Conversation skills during generalization probes, with Perspective Taking demonstrating improved, yet variable, demonstration. Mean levels of skill accuracy during maintenance ranged from 90% to 100% in the training setting and from 77% to 100% in the generalization setting. NAP calculations for intervention phase generalization probes of Participation and training setting probes of Conversation during maintenance phase revealed moderate effects (Table 2). All other NAP calculations were interpreted as strong effects.

During baseline, Sloane demonstrated declining accuracy of the Participation skill, variable accuracy of the Conversation skill, and low accuracy of the Perspective Taking and Problem Solving skills during baseline in both the training and generalization settings (Fig. 2). Mean levels of accuracy during baseline ranged from 33% to 72% for training setting probes and from 5% to 77% for generalization probes. Immediate increases in level and trend from baseline to intervention were observed for the Participation, Conversation, Perspective Taking, and Problem Solving skills in both the training and generalization settings. Mean levels of skill accuracy during intervention ranged from 93% to 100% in the training setting and from 81% to 100% in the generalization setting. Additionally, decreases in variability were observed for all target skills. During maintenance probes, mean levels of skill accuracy were found to be 100% for all skills in both training and generalization settings. Calculation of NAP revealed strong effects for all probes in the training setting with the exception of the Participation skill during the intervention phase, which is interpreted as a moderate effect (Table 2). NAP calculations revealed moderate generalization effects for Participation and Conversation skills, with strong effects observed for Perspective Taking and Problem Solving skills.

Ferris demonstrated variable accuracy of the Participation skill during baseline, with low levels of skill accuracy observed for the Conversation, Perspective Taking, and Problem Solving skills (Fig. 3). Mean levels of skill accuracy during baseline ranged from 5% to 52% in the training setting and from 2% to 47% in the generalization setting. Following the introduction of social skills training, immediate increases in level and trend were observed for all skills in both the training and generalization settings. Mean levels of skill accuracy during intervention ranged from 92% to 97% in the training setting and from 33% to 88% in the generalization setting. Maintenance probes in both the training and generalization settings found Participation, Conversation, and Perspective Taking skills use to be improved from baseline levels. Mean levels of skill accuracy during maintenance ranged from 85% to 97% for training setting probes and from 56% to 100% for generalization probes. NAP calculations for generalization probes of the Problem Solving skill revealed a moderate effect, with all other skills demonstrating strong effects (Table 2).

Table 2
Skill Acquisition and Generalization During Intervention and Maintenance.

	Cameron		Sloane		Ferris	
	Treatment	Generalization	Treatment	Generalization	Treatment	Generalization
Participation-Intervention	**1.00** (0.48–1.51)	0.86 (0.21–1.24)	0.83 (0.15–1.18)	0.89 (0.26–1.30)	**0.98** (0.45–1.48)	**0.97** (0.42–1.46)
Participation-Maintenance	**1.00** (0.59–1.41)	**1.00** (0.59–1.41)	**0.94** (0.48–1.30)	0.89 (0.37–1.19)	**0.94** (0.47–1.29)	**1.00** (0.59–1.41)
Conversation-Intervention	**0.97** (0.44–1.42)	**0.99** (0.48–1.46)	**0.92** (0.35–1.32)	0.83 (0.18–1.16)	**1.00** (0.51–1.49)	**0.98** (0.47–1.45)
Conversation-Maintenance	0.90 (0.40–1.19)	**0.96** (0.52–1.31)	**0.92** (0.44–1.23)	0.83 (0.27–1.01)	**1.00** (0.60–1.40)	**0.99** (0.60–1.29)
Perspective Taking-Intervention	**1.00** (0.59–1.41)	**0.96** (0.50–1.32)	**1.00** (0.57–1.43)	**0.93** (0.43–1.30)	**1.00** (0.59–1.41)	**0.93** (0.44–1.46)
Perspective Taking-Maintenance	**1.00** (0.59–1.41)	**0.99** (0.63–1.33)	**1.00** (0.57–1.43)	**1.00** (0.57–1.43)	**1.00** (0.59–1.41)	**1.00** (0.59–1.41)
Problem Solving-Intervention	**1.00** (0.59–1.41)	**1.00** (0.59–1.41)	**1.00** (0.59–1.41)	**1.00** (.0.59–1.41)	**1.00** (0.59–1.41)	0.86 (0.37–1.07)

Note. NAP effect size scores below .66 are considered small, scores between .66 and .92 are considered moderate, and scores ranging from .92 to 1.0 are considered strong (Parker & Vannest, 2009). Strong effects in **bold**. Ninety percent confidence intervals in parentheses.

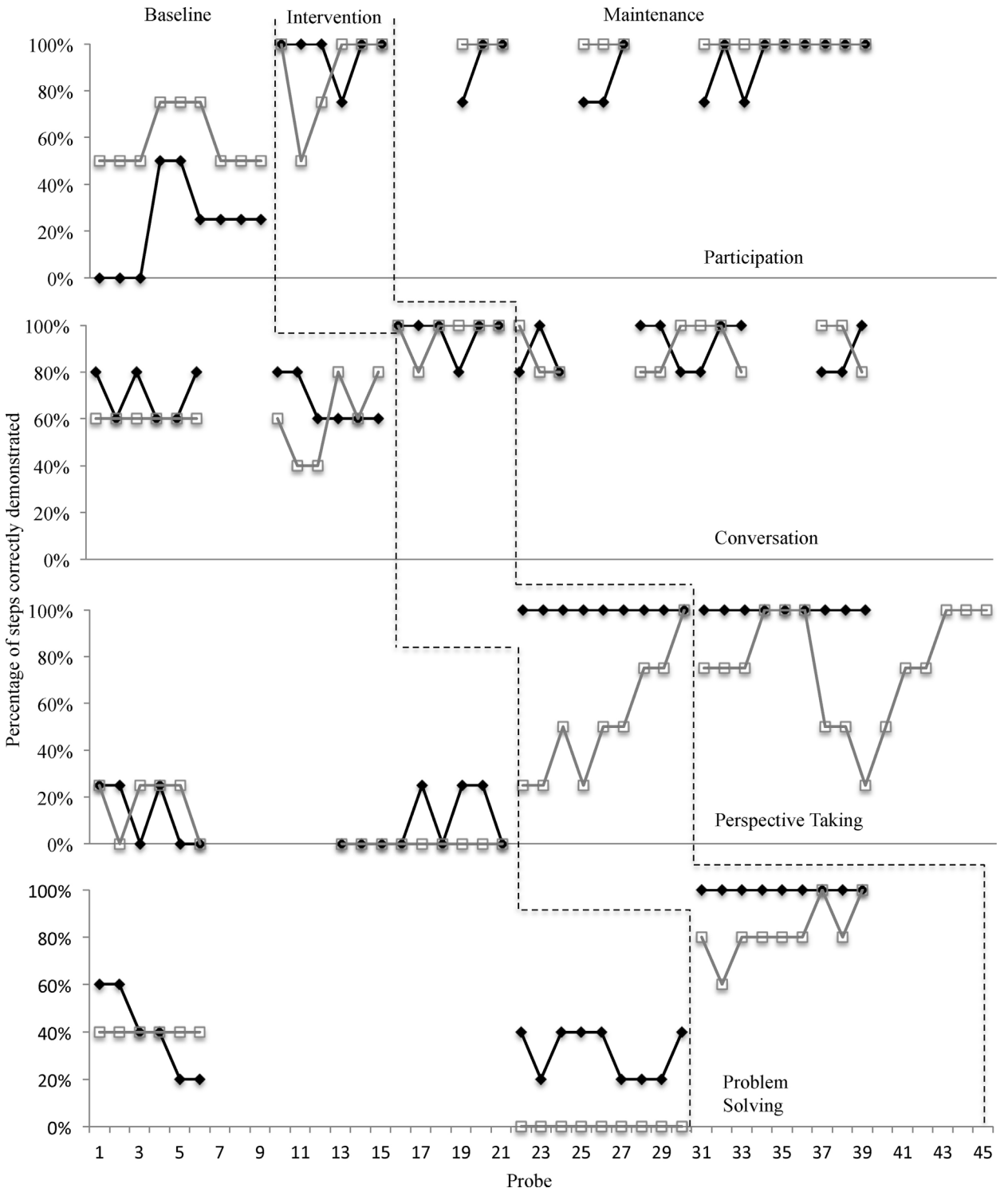


Fig. 1. Percentage of steps correctly demonstrated in treatment and generalization settings, Cameron. Closed diamonds represent training setting probes, open squares represent generalization setting probes.

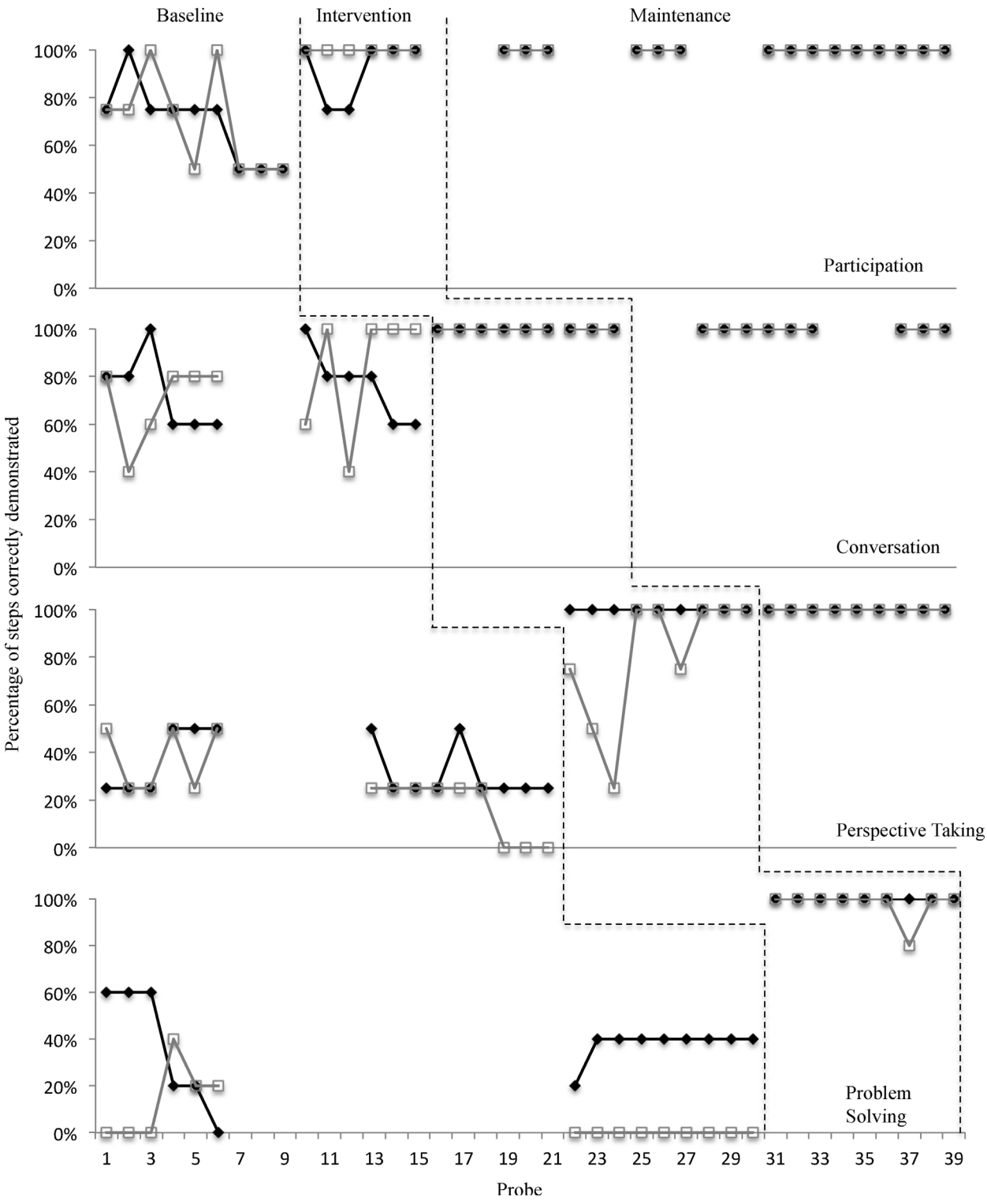


Fig. 2. Percentage of steps correctly demonstrated in treatment and generalization settings, Sloane. Closed diamonds represent training setting probes, open squares represent generalization setting probes.

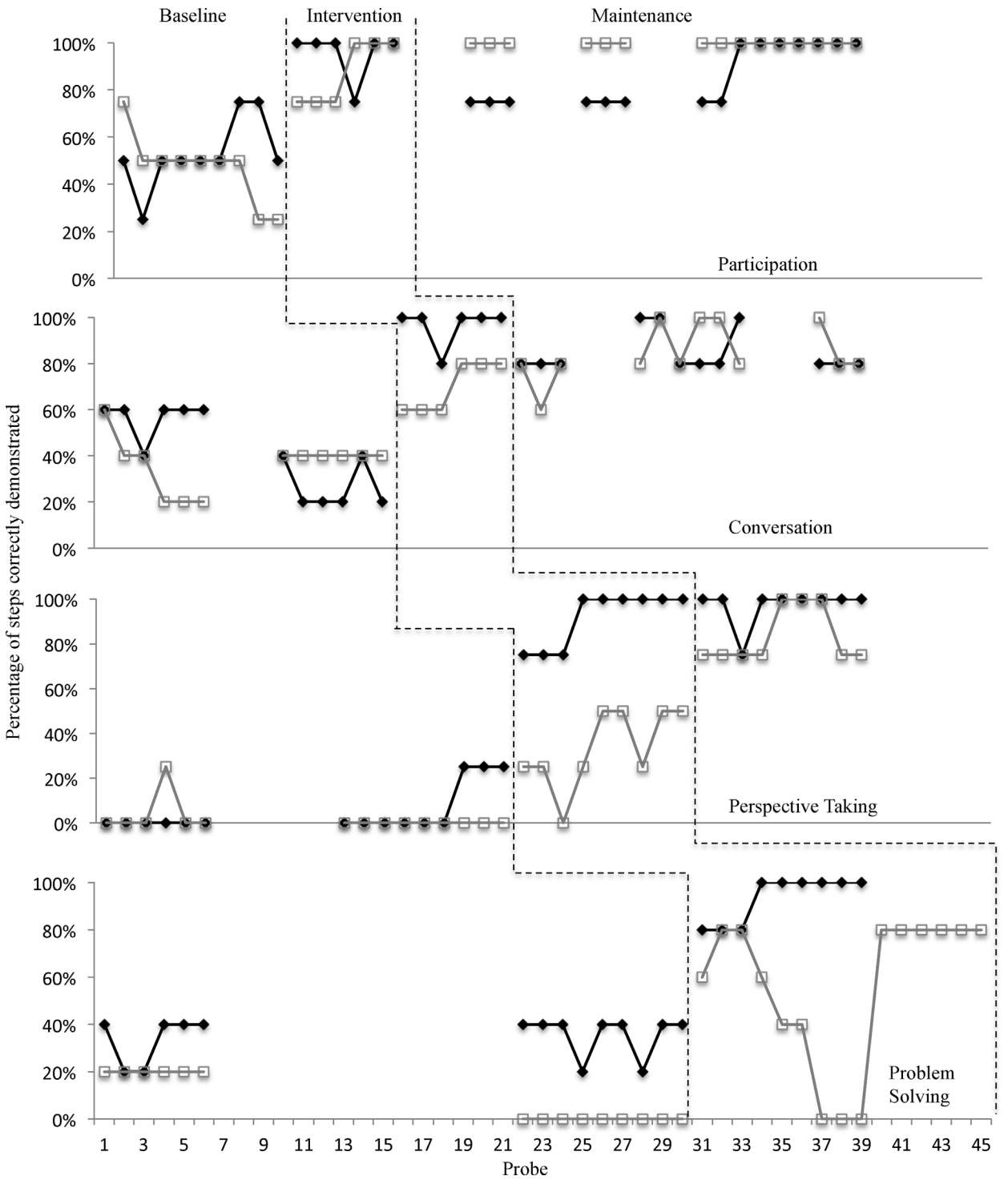


Fig. 3. Percentage of steps correctly demonstrated in treatment and generalization settings, Ferris. Closed diamonds represent training setting probes, open squares represent generalization setting probes.

Table 3
Pre- and post-intervention ASSP scores.

Scale	Cameron		Sloane		Ferris	
	Pre	Post	Pre	Post	Pre	Post
Social Reciprocity	56	61	39	40	37	39
Detrimental Social Behaviors	27	29	23	23	25	23
Participation/Avoidance	33	36	24	25	24	27
Total	127	137	94	97	95	95

Table 4
Pre- and post-intervention PSI/SF scores.

Scale	Cameron		Sloane		Ferris	
	Pre	Post	Pre	Post	Pre	Post
Parental Distress	26	24	34	26	36	33
Parent–Child Dysfunctional Interaction	27	25	33	32	44	37
Difficult Child	32	26	40	37	52	49
Total Stress	85	75	107	95	132	119

2.2. Social functioning

Social functioning of participants was assessed though a parent- or guardian-completed ASSP (Bellini & Hopf, 2007). Pre- and post-intervention ASSP results for individual participants are presented in Table 3. Improvements were observed for Cameron and Sloane, with no improvement reported in Ferris' Total Social Functioning score. Paired samples *t*-tests were conducted on the group ASSP data. Results revealed that improvements were not considered significant for any of the subscales, including the Social Reciprocity subscale, $t(2) = 2.22$, $p = .16$, $r = .73$, the Participation/Avoidance subscale, $t(2) = 3.50$, $p = .07$, $r = .80$, the Detrimental Social Behaviors subscale, $t(2) = 0.00$, $p = 1.00$, $r = .00$, and the Total Social Functioning score, $t(2) = 1.46$, $p = .28$, $r = .65$.

2.3. Parental stress

Parental stress levels were measured for the parent or guardian of each participant using the PSI/SF (Abidin, 1995). Pre- and post-intervention Parental Distress (PD), Parent–Child Dysfunctional Interaction (P-CDI), Difficult Child (DC), and Total Stress scores for each parent are presented by participant in Table 4. Overall, clinically large reductions were observed for each parent. Paired samples *t*-tests were conducted on the group data for each category. A significant and clinically meaningful reduction was found in Total Stress, $t(2) = 13.23$, $p = .01$, $r = .99$) The reduction in the area of DC was clinically large and approaching significance, $t(2) = 4.00$, $p = .06$, $r = .94$. The reductions in the areas of PD, $t(2) = 2.34$, $p = .15$, $r = .86$, and P-CDI, $t(2) = 1.80$, $p = .21$, $r = .79$, were not significant.

2.4. Social validity

At the conclusion of the intervention, parents and guardians completed the BIRS (Elliott & Treuting, 1991) to assess acceptability of the intervention. Possible BIRS scores ranged from 1 to 6, with higher scores indicating a greater level of acceptability. Results of the BIRS indicate moderate to high levels of social validity of the intervention. Overall item mean for both Cameron's and Ferris' parents were 4.96, and item mean for Sloane's guardian was 4.50.

3. Discussion

The current study evaluated whether participation in the Superheroes Social Skills program would result in improvements in accurate demonstration of target skills within a training setting. Participants in the study demonstrated immediate increases in level and trend of skill use, as well as moderate to strong effect sizes in improving demonstration of skills from baseline to intervention phases in the training setting. As such, these results were consistent with prior research that has found social skills training to improve accurate skill demonstration in training settings (e.g., Ganz et al., 2012). Probes of skill maintenance revealed that skills were maintained at levels similar to intervention following withdrawal of direct skill instruction. This finding is notable, as previous research has suggested limited maintenance effects of social skills training (Bellini et al., 2007). Maintenance effects observed in the present study may be attributed to inclusion of intervention strategies such as video modeling, previously found to be beneficial for increasing skill maintenance in children with ASD (Wang & Spillane, 2009).

The current study also evaluated the effect of participation in the social skills intervention on generalization of skills across persons, settings, and stimuli. Data collected suggest that the participants generalized use of targeted social skills. Immediate increases in level and trend from baseline to intervention were observed for all of the skills during generalization probes, with improvements representing moderate to strong effects. Although generalization of target skills through social skills training is often limited (Bellini et al., 2007; Rao et al., 2008), generalization was observed in the current study. Generalization of target skills may be due to exploitation of current functional contingencies, training diversely, and incorporation of functional mediators of behavior, as Stokes and Osnes (1989) identified these strategies as promoting generalization of skills. Additionally, utilization of sequential modification (Stokes & Baer, 1977) likely contributed to generalization of target social skills. Similar to probes in the training setting, maintenance at or above intervention levels was observed for all participants during generalization probes.

Skill use in both training and generalization settings in the current study represent meaningful improvements, as the social skills targeted were reported to be salient social deficits. Importantly, the greatest improvements in skill accuracy were noted in Perspective Taking and Problem Solving, which represent more complex social skills (e.g., Rao et al., 2008). However, it is likely that the smaller improvements in the Participation and Conversation skills were due to ceiling effects associated with higher entry accuracy of these skills. Although the current study suggests that the Superheroes Social Skills program was effective in promoting improvements in accurate skill demonstration in training and generalization settings for each targeted skill, further analysis is warranted to investigate the utility of other skills included in the curriculum for promoting use of target skills.

Although a paired samples *t*-tests of parent-completed ASSP data did not find statistically significant improvements, both Cameron and Sloane demonstrated improvements in Total Social Functioning scores. Despite demonstrating improvements in accurate skill demonstration during training and generalization probes, Ferris' Total Social Functioning score did not change over the course of intervention. This may have occurred for several reasons. It is important to note that Ferris demonstrated improvements on the Social Reciprocity and Participation/Avoidance subscales, while the Detrimental Social Behaviors score was endorsed as declining. As Ferris was reported to engage in aggressive and disruptive behaviors, as well as other behaviors that may limit social success in home and community settings (e.g., toileting accidents), it is likely that these behaviors contributed to a lack of improvement on the ASSP Total Social Functioning score.

Improvements on the PSI/SF also suggest that participants may have generalized skill use to non-training environments. Clinically meaningful and statistically significant reductions were observed in the areas of Total Stress and Difficult Child. Although Radley et al. (2014) found parent stress to be reduced when parents were trained to facilitate the Superheroes Social Skills program, results of the current study suggest that child participation alone may be effective in improving levels of parental stress. This finding is noteworthy as levels of stress in parents of children with ASD have been found to be stable over time (Lecavalier, Leone, & Wiltz, 2006), suggesting that improvements in target social skill use may have contributed to improved participant behavior and decreased parent stress in non-training settings (e.g., home).

Although the current study has implications for applied practice, results must be viewed in light of several limitations. First, the Superheroes Social Skills program incorporates multiple strategies to promote acquisition and generalization of social skills. The inclusion of multiple generalization-promoting strategies limits the ability to distinguish which of the components was functionally related to behavioral change in training and generalized settings. Future research should seek to address this limitation through conducting a component analysis of the Superheroes Social Skills program. Second, due to the group format of social skills instruction, training for the four target social skills occurred in the same order for all participants. As such, it is unknown whether certain target skills are prerequisite for acquisition of other target skills, or whether alterations in training sequence may affect acquisition of target skills. Future research may address this limitation through counterbalancing treatment sequence across multiple groups. Third, maintenance data were collected over approximately two weeks. As such, conclusions regarding long-term maintenance of acquired social skills are limited. Future research should seek to address this limitation though collection of maintenance data over several months. Although parent-competed ASSP and PSI/SF suggest improvements in child behavior in home and community settings, no direct measures of skill use were collected in these settings. Future research may address this limitation through evaluation of skill use in these environments. The current study also did not assess procedural integrity of parents in reviewing videos of role-plays or reading social narrative comics. Future research should collect parent procedural integrity data. Lastly, each of the participants demonstrated at least one step of each target skill during baseline. As such, it is unclear whether similar results would be observed in participants who did not exhibit any steps of target skills prior to training.

4. Conclusion

Although social skills training programs are often found to promote skill acquisition in training settings (e.g., Plavnick et al., 2013), effects often fail to generalize to non-training settings (Bellini et al., 2007). Given that social competence is dependent on successfully utilizing social skills in context, social skills training programs must place an emphasis on programming for skill generalization. Results of the current study suggest that the Superheroes Social Skills program, through incorporation of multiple strategies to encourage generalization, was successful in promoting generalized skill use. Although future research is needed to address limitations of the current study, results suggest that the Superheroes Social Skills program may be useful to practitioners as a means of improving social competence of children with ASD.

Appendix A

Treatment Integrity Checklist

Superheroes Social Skills program components	Y/N
1. Check-in. Have materials placed in front of the group; children are seated; brief review of prior session (where applicable)	
2. Review rules and purpose of session	
3. Introduce skill and explain rationale	
4. Watch corresponding fast hands and video models	
5. Facilitator role-plays both a positive and negative example of the skill	
6. Assess skill accuracy in training setting	
7. Unrestricted free time	
8. Assess skill accuracy in generalization setting	
9. Watch animated social narrative comic book	
10. Provide reward for participation	
11. Encourage participants to watch recording of role-play and read social narrative comic book	

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